SUSTAINABLE MAIZE FARMING INITIATIVE

(Phase 1 – Starting 2026-2027 rain fed farming season)

Submitted By: Projects Committee - Adventist Youth Ministries (AY), Woodlands Conference

1. INTRODUCTION AND EXECUTIVE SUMMARY

The Adventist Youth Ministries (AY) of the Woodlands Conference presents this fully-funded Phase 1 pilot project for a sustainable commercial maize farming venture.

With seed capital of ZMW 200,000 already available in the conference bank account, this initiative will acquire farmland (8-10 hectares) and establish maize production. The remaining land is reserved for future expansion.

IMPORTANT FINANCIAL PRINCIPLE:
• The K200,000 seed capital is ONLY for land, water, solar power, and maize farming.
• Livestock and crop diversification will be funded EXCLUSIVELY from maize revenue profits.
• Maize profits will first cover next season's farming costs, then fund expansion, then grow bank savings.
• Fencing, livestock enclosures, and title processing will be done in future years using profits.

The dual mission remains:
1. Generate sustainable income to fund evangelistic campaigns and community support programmes.
2. Empower youth through practical agribusiness.

2. SEED CAPITAL BUDGET (ZMW 200,000)

Item	Cost (ZMW)
1. LAND ACQUISITION	
Farmland (8-10 hectares)	40,000
2. WATER INFRASTRUCTURE	
Borehole Drilling	15,000

Submersible Pump	10,000
Water Pipes & Fittings	3,000
Water Tank (1 × 5,000L)	6,000
3. SOLAR POWER SYSTEM	
Solar Panels (4 × K2,000 = 1.4kW system)	8,000
Solar Battery (Deep Cycle, smaller)	6,000
Inverter (1.5kW Pure Sine Wave)	8,000
Cabling	2,000
Installation & Labour	1,000
4. MAIZE FARMING INPUTS (8 hectares)	
Tractor Hire – Ploughing (8 ha × K1,000)	8,000
Maize Seed – 10kg bag per hectare (8 × K600)	4,800
Basal Fertilizer – D Compound (16 × 50kg bags @ K800)	12,800
SUBTOTAL (Fixed Costs)	168000
OPERATING RESERVE (Kept in bank)	32000
TOTAL SEED CAPITAL	**200,000**

Note: All K200,000 is readily available. No external funding required.

3. LAND USE PLAN (8-10 HECTARES)

• Total land: 8-10 hectares at ZMW 40,000

• Phase 1 allocation (Year 1–2):

- 8 hectares: Rain-fed maize production

- Remaining land: Reserved for future livestock and crop diversification

• Water tank (1 × 5,000L) stores water for irrigation.

• Solar power system (1.4kW) provides electricity for water pump year-round.

• Fencing and enclosures will be added in future years using maize profits.

4. MAIZE PRODUCTION – 8 HECTARES (Year 1)

4.1 Rain-fed Maize Only (First Year)

• Cultivated area: 8 hectares during rainy season.

• Expected yield: 30 × 50kg bags per hectare (conservative estimate).

• Total bags: 8 × 30 = 240 bags.

• FRA price: K350 per 50kg bag.

• Revenue: 240 × K350 = ZMW 84,000.

Note: Winter green maize will begin in Year 2 after solar-powered borehole and water tank are fully operational.

4.2 Year 1 Net Profit Calculation

Total Maize Revenue (8 ha, 240 bags @ K350)	K84,000
Less: Tractor Hire – Ploughing (8 ha @ K1,000)	K8,000
Less: Maize Seed (8 × K600)	K4,800
Less: Fertilizer (D Compound + Urea)	K26,400
Less: Herbicides/Pesticides	K4,000
Less: Labour (6 months @ K1,000 + Casual)	K17,000
Less: Transport	K4,000
Less: Post-Harvest (Shelling + Bagging)	K4,800
TOTAL PRODUCTION COSTS	K69,000
NET PROFIT (Year 1)	**K15,000**

This K15,000 net profit will be used for:

- Next season's farming inputs (revolving fund)
- Starting winter green maize production (Year 2)
- Small savings to grow bank balance

5. SOLAR POWER SYSTEM DETAILS

The solar power system ensures year-round water pumping without reliance on grid electricity or fuel generators.

System Components:

- Solar Panels: 1.4kW system (4 panels × K2,000 each) – generates ~6-8 kWh per day
- Solar Battery: Deep cycle battery – stores energy for night/cloudy days (K6,000)
- Inverter: 1.5kW Pure Sine Wave – converts DC to AC for the submersible pump (K8,000)
- Cabling & Installation: K3,000 total
- Water Tank: 1 tank (K6,000) for water storage

Benefits:

- Zero fuel costs after installation
- Reliable water supply for irrigation
- Low maintenance
- Environmentally friendly
- Water storage ensures supply during cloudy days

6. YEAR 2 – WINTER MAIZE (Green Cobs)

After a successful Year 1 rain-fed harvest, profits will fund winter maize production on 4 hectares using the solar-powered borehole, pump, and water tank.

- Area: 4 hectares of green maize
- Expected yield: 40,000 cobs per hectare
- Total cobs: 4 × 40,000 = 160,000 cobs
- Selling price: K5 per cob
- Revenue: 160,000 × K5 = K800,000

Year 2 Production Costs (green maize only):

- Tractor hire (4 ha ploughing): K4,000
- Seed (4 × K600): K2,400
- Fertilizer (reduced for green maize): K12,000
- Labour (12 months @ K1,000): K12,000
- Total Year 2 costs: ~K30,400 (no fuel costs – solar powered)

Year 2 Net Profit (Rain-fed 8 ha + Green 4 ha):

- Rain-fed revenue: K84,000 (as Year 1)
- Green maize revenue: K800,000

- Total revenue: K884,000
- Total costs (both): K69,000 + K30,400 = K99,400
- NET PROFIT Year 2: K784,600

7. LIVESTOCK EXPANSION (Year 3 onwards – Funded by Maize Profits)

No livestock will be purchased using the K200,000 seed capital.

Livestock will only be introduced after maize profits have:

1. Covered all farming operational costs
2. Built a savings buffer in the bank account
3. Provided surplus for investment

The solar-powered borehole and water tank will provide water for all livestock.

Fencing and enclosures will be funded from maize profits before introducing livestock.

7.1 Village Chickens (Year 3)

- Start with 50 chicks (cost ~K500 from maize profits)
- Incubator machine: K3,500 (one-time purchase)
- Selling price: K200 per mature bird after 6 months
- Rapid flock growth using incubator

7.2 Goats (Year 3–4)

- Start with 5 does + 1 buck (cost ~K3,600 from maize profits)
- Each doe gives birth to 2 kids per year
- Selling price: K600 per mature goat

7.3 Cattle (Year 4–5)

- Start with 2–3 young cows (K3,000 each) + 1 bull
- After 2–3 years, sell mature cows for K10,000+
- Funded entirely from accumulated maize profits

8. PROJECTED BANK SAVINGS GROWTH

Year	Activity	Bank Balance (End of Year)
Year 0	Seed capital after all purchases	K43,000 (reserve)
Year 1	Rain-fed maize only – net profit K15,000	K58,000
Year 2	Rain-fed + green maize – net profit K784,600	K842,600
Year 3	Maize + chickens – net profit ~K850,000	K1,692,600
Year 4	Maize + chickens + goats – net profit ~K950,000	K2,642,600
Year 5	Maize + all livestock + diversification	K3,800,000+

9. SUSTAINABILITY AND REINVESTMENT PRINCIPLES

1. Seed capital (K200,000) is NEVER used for livestock – only for land, water, solar, maize.

2. Solar power eliminates fuel costs for water pumping.

3. Water tank provides storage for irrigation.

4. Maize revenue pays for next season's farming inputs FIRST.

5. Surplus maize profits fund livestock expansion and fencing gradually.

6. Bank savings grow year by year – never depleted.

10. CONCLUSION

With ZMW 200,000 already in the bank, the Woodlands Conference AY Projects Committee is ready to implement Phase 1 immediately.

This plan follows strict financial discipline:
• Seed capital ONLY for land, water, solar power, and maize farming
• K40,000 for 8-10 hectares of farmland
• Solar panels, battery, inverter, and water tank for year-round water pumping
• 8 hectares of rain-fed maize in Year 1
• Winter green maize in Year 2 funded by Year 1 profits
• Livestock introduced in Year 3+ funded by maize profits
• Fencing and title processing done in future years using profits
• Bank savings grow continuously

DETAILED MAIZE FARMING BUDGET

A. LAND ACQUISITION

Item	Unit Cost (ZMW)	Total (ZMW)
Farmland (8-10 hectares)	Lump sum	40,000

B. WATER INFRASTRUCTURE

Item	Unit Cost (ZMW)	Total (ZMW)
Borehole Drilling (40-60m depth)	Lump sum	15,000
Submersible Pump	Lump sum	10,000
Water Pipes & Fittings	Lump sum	3,000
Water Tank (1 × 5,000L)	Lump sum	6,000

C. SOLAR POWER SYSTEM (For Water Pump)

Item	Unit Cost (ZMW)	Total (ZMW)
Solar Panels (4 × 350W = 1.4kW system)	2,000 per panel	8,000
Solar Battery (Deep Cycle Battery)	Lump sum	6,000
Inverter (1.5kW Pure Sine Wave)	Lump sum	8,000
Cabling	Lump sum	2,000
Installation & Labour	Lump sum	1,000

D. LAND PREPARATION (8 hectares)

Item	Unit Cost (ZMW)	Total (ZMW)
Tractor Hire – Ploughing (8 hectares)	1,000 per hectare	8,000
Manual Labour – Clearing & Bunds (8 hectares)	500 per hectare	4,000

E. MAIZE SEED

Item	Unit Cost (ZMW)	Total (ZMW)
Certified Maize Seed (8 hectares)	600 per 10kg bag	4,800

F. FERTILIZER

Item	Unit Cost (ZMW)	Total (ZMW)
Basal Fertilizer – D Compound (16 × 50kg bags)	800 per 50kg bag	12,800
Top Dressing – Urea (16 × 50kg bags)	850 per 50kg bag	13,600
Application labour	100 per hectare	800

G. HERBICIDES & PESTICIDES

Item	Unit Cost (ZMW)	Total (ZMW)
Pre-emergence herbicide (Glyphosate)	400 per litre × 4 litres	1,600
Post-emergence herbicide	350 per litre × 3 litres	1,050
Insecticide (Fall Armyworm control)	300 per litre × 2 litres	600
Spraying labour	50 per hectare	400

H. TRANSPORT (Lusaka to Farm - 150km)

Item	Unit Cost (ZMW)	Total (ZMW)
Transport of all equipment, inputs, and labour	Lump sum	4,000

I. LABOUR COSTS (6 months – Rain-fed Season)

Item	Unit Cost (ZMW)	Total (ZMW)
Farm Worker/Caretaker (6 months)	1,000 per month	6,000
Casual labour – Planting (8 days × 5 workers)	100 per day	4,000
Casual labour – Weeding (8 days × 5 workers)	100 per day	4,000
Casual labour – Harvesting (10 days × 5 workers)	100 per day	5,000

J. POST-HARVEST

Item	Unit Cost (ZMW)	Total (ZMW)
Shelling (240 bags)	15 per bag	3,600
Bagging (240 bags)	5 per bag	1,200

GRAND TOTAL SUMMARY – YEAR 1 MAIZE (8 HECTARES)

A. Land Acquisition	K40,000
B. Water Infrastructure	K34,000
C. Solar Power System	K25,000
D. Land Preparation	K12,000
E. Maize Seed	K4,800
F. Fertilizer	K27,200
G. Herbicides & Pesticides	K3,650
H. Transport	K4,000
I. Labour (6 months + casual)	K19,000
J. Post-Harvest	K4,800
TOTAL FIXED & OPERATING COSTS	**K174,450**

IMPORTANT NOTES:

• The total above (K174,450) represents the full value of assets and costs.

• From the K200,000 seed capital, the breakdown is:

- K174,450 spent on land, water, solar, inputs, labour, transport, and post-harvest
- K25,550 kept as operating reserve in the bank

• Expected revenue: 240 bags × K350 = K84,000.

• Year 1 Net Profit: K84,000 - K69,000 (production costs) = K15,000.

• Year 2 green maize(cobs) (4 hectares) will generate K800,000 revenue to recover investment.